AWS VPC Peering Internship Project

EWB Tech Academy | Secure Cross-VPC Networking Architecture

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1. Project Architecture Overview

This project establishes a secure network bridge between two isolated Virtual Private Clouds (VPCs) in AWS using **VPC Peering**. This architecture enables private, low-latency communication without routing traffic over the public internet.

- **VPC A (`10.10.0.0/16`)**: Configured with a public subnet (`ewb-a-public`, `10.10.1.0/24`) housing a Bastion Host (`ewb-bastion`), and a private subnet (`ewb-a-private`, `10.10.2.0/24`) hosting application server `ewb-app-a`.
- **VPC B (`10.20.0.0/16`)**: Configured with a private subnet (`ewb-b-private`, `10.20.1.0/24`) hosting application server `ewb-app-b`.
- **Peering Connection (`ewb-a-b-peer`)**: Establishes bidirectional connectivity between VPC A and VPC B (`pcx-0f0d76bc8799945b3`).

> **Design Objective:** Direct instance-to-instance private routing secured via explicit security group rules and custom route table associations.

2. Resource Inventory

Resource Type	Name Tag	IP / CIDR Block	Location / VPC
VPC A	`ewb-vpc-a`	`10.10.0.0/16`	Mumbai (`ap-south-1`)
VPC B	`ewb-vpc-b`	`10.20.0.0/16`	Mumbai (`ap-south-1`)
Public Subnet	`ewb-a-public`	`10.10.1.0/24`	`ewb-vpc-a`
Private Subnet A	`ewb-a-private`	`10.10.2.0/24`	`ewb-vpc-a`
Private Subnet B	`ewb-b-private`	`10.20.1.0/24`	`ewb-vpc-b`
Bastion Instance	`ewb-bastion`	Public IP / `10.10.1.10`	`ewb-a-public`
App Instance A	`ewb-app-a`	`10.10.2.10` (Private)	`ewb-a-private`
App Instance B	`ewb-app-b`	`10.20.1.10` (Private)	`ewb-b-private`
Peering Link	`ewb-a-b-peer`	`pcx-0f0d76bc8799945b3`	VPC A ⟷ VPC B

3. Verification & Testing Evidence

To validate the cross-VPC peering infrastructure, connectivity tests were performed using ICMP echo requests between private application servers.

- **Forward Ping Test (`ewb-app-a` $\rightarrow$ `ewb-app-b`):** Executed `ping -c 4 10.20.1.10` from `ewb-app-a`.
- **Result:** 4 packets transmitted, 4 received, **0% packet loss** with an average round-trip time of ~0.36 ms.
- **Security & Routing:** Confirmed that private subnets correctly route traffic through the active peering attachment based on destination CIDR blocks (`10.20.0.0/16` and `10.10.0.0/16`).

> **Conclusion:** The VPC Peering project was successfully deployed, tested, and documented in accordance with EWB Tech Academy internship standards.